



Do New Economic Geography agglomeration shadows underlie current population dynamics across the urban hierarchy?*

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Received: 21 September 2007 / Accepted: 17 July 2008

Abstract. Although the New Economic Geography (NEG) has been used extensively to formally explain the emergence of the American urban system, few studies investigate its success in explaining current population dynamics in a more established urban system. This study explores whether proximity to same-sized and higher-tiered urban centres affected the patterns of 1990–2006 US county population growth. Rather than casting NEG agglomeration shadows on nearby growth, larger urban centres generally appear to have positive growth effects for more proximate places of less than 250,000 people. However, there is some evidence the largest urban areas cast growth shadows on proximate medium-sized metropolitan areas and of spatial competition among small metropolitan areas.

JEL classification: R11, R12

Key words: New Economic Geography, agglomeration shadow, population growth, urban hierarchy

1 Introduction

The interrelationships among urban centres and between them and their rural fringes are among the most visible features of the expanding American urban landscape. Responding to technological, economic, and quality-of-life stimuli, households relocate to areas that offer greater net utility. Firms' cost-minimizing location decisions simultaneously influence household decisions and also respond to them. The resulting population flows drive the evolution of the hierarchical urban system.

* We thank Jordan Rappaport and Taisuke Nakata of the Federal Reserve Bank of Kansas City for generously providing Stata codes for the generalized method of moments (GMM) estimation. We appreciate comments we received at presentations at Central Florida University, Oklahoma State University, the 45th Annual Meetings of the Southern Regional Science Association in St. Augustine, FL, and the Transatlantic Land Use Conference in Washington, DC.

Interest in the spatial dimension of population dynamics burgeoned with the advent of the New Economic Geography (NEG) (Krugman 1991), which built on the urban–hierarchy lattice from traditional central place theory (CPT) (Christaller 1933). In CPT, lower-tiered places depend on higher-tiered places for access to progressively higher-ordered goods and services offered at each tier. In extending CPT, the NEG formalizes the role of agglomeration in the dynamic formation of an urban system. Both theories prominently feature an urban hierarchy based on regional market potential, creating symbiotic interrelationships among tiers, including the rural fringe (Fujita et al. 1999). A core prediction of standard NEG formulations is the existence of agglomeration shadows preventing urban areas from forming too closely to other equal or larger-size urban areas due to fierce spatial price competition.

Explaining the emergence of the American urban system through the early 20th Century was one of the primary goals of NEG proponents. Despite significant successes in this area, there is a need for better empirical understanding of how agglomeration economies influence *current* population mobility (Eeckhout 2004; Tabuchi and Thisse 2006). For example, as a result of the declining costs of transporting goods relative to moving people, NEG models may less accurately predict 21st century US population movements (Glaeser and Kohlhase 2004). Further, while growth shadows may define the constraints in terms of equal or higher-tier places emerging next to an urban centre, the relationships with smaller tiered places and the rural hinterlands may be quite different.

Most related research is limited to larger urban centres and metropolitan areas (MAs) (Eaton and Eckstein 1997; Glaeser and Maré 2001; Lucas and Rossi-Hansberg 2002; Head and Mayer 2004, 2006; Brühlhart and Koenig 2006; Knaap 2006) with surprisingly little work examining spatial interactions beyond major urban areas.¹ Partridge et al. (2008b) examined population growth for nonmetropolitan and small metropolitan areas for 1950–2000 within the context of the urban hierarchy. However, they did not consider interactions among mid-sized and larger metropolitan areas, particularly the presence of potential agglomeration shadows on growth, and did not focus on the most recent growth experience. Recent changes in transportation costs (Glaeser and Kohlhase 2004), technology, and the nature of transactions costs and firm interactions (McCann and Shefer 2004; McCann 2007), may have caused shifts in spatial growth dynamics. Continued policy interests, such as land use management, underscore the need to understand the current general applicability of NEG models for growth within the urban hierarchy.

One implication of NEG (CPT) models is that spatial price competition may limit the number of firm births and expansions in areas nearby large urban centres. This could translate into population growth being positively related to distance from (other) similar-sized or larger urban centres; namely, there are ‘growth shadows’. The landscape of growth in the vicinity of urban centres, due to the offsetting effects of positive agglomeration spillovers and spatial price competition is, however, not well-understood. With both positive spillovers and ‘growth shadow’ effects likely present, the net impact on population growth will in part reflect which one of these two effects is greater.

Agglomeration economies may have a greater *positive* geographic reach than is usually assumed, with cities being an economic driver far beyond their borders (Partridge et al. 2008b). The relationship between lower- and higher-tiered places is increasingly comprised of commuting to (urban) jobs, accessing higher-ordered urban services and amenities, and urban sprawl and related centrifugal forces that push residences further into rural exurbia. Improved transportation and communication technologies may have extended spatial urban input–output

¹ One interesting exception is Stelder (2005) whose simulations examined whether the European urban hierarchy can be explained by a NEG model when starting from a uniform population distribution. Though there are some successes, NEG has a difficult time explaining ‘clusters’ of cities such as the German Ruhrgebiet.

linkages beyond what is typically presumed. The range of goods and services for which spatial competition limits growth will also be closely related to the tier of the nearby urban centre. To reveal the spatial richness that underlies agglomeration spillovers and potential growth shadows in the urban system, distances and interactions among the full set of proximate higher-tiered urban centres must be directly considered.

This study fills this void by using both 1990–2000 and 1990–2006 US county level population growth, which allows us to consider interactions among places from the lowest to highest-tiered urban centres. Our conceptual departing point from NEG/CPT follows from a model of city development, in which we allow for the possibility of commuting and input–output externalities outside of the ‘city’ border, along with the desire for proximity to urban amenities and services successively available in larger urban centres. With these extended commuting or input–output linkages, agglomeration spillovers may prevail over a broader geographic area, offsetting or eclipsing growth shadow effects. Population growth in areas nearby similar-sized or larger urban areas may be *inversely* related to distance from the city rather than positively related (i.e., growth shadows) as predicted by standard NEG/CPT models.

Specifically, we focus on how an individual county’s geographic location in the urban hierarchy affects its growth, based on distances to small urban centres all the way up to the largest centres. We examine whether nearby similar-sized or larger urban neighbours on net provide beneficial access to agglomeration effects, or produce a growth shadow as predicted in many NEG/CPT formulations. We conclude that some elements of NEG models appear valuable in explaining the evolution and interactions within a more established urban system, but other perspectives and models also are needed.

2 Theoretical considerations

The distribution of economic activity across the American urban landscape reflects a multitude of location decisions by firms and households. A synthesis of theories regarding how these decisions cause population to agglomerate into core areas informs our empirical model and distinguishes the key competitors from the NEG (CPT) framework (Brühlhart 1998).

Traditional neoclassical economic theory explains an uneven spatial distribution of economic activity as resulting from regional differences in geography, endowments, and exogenous technology. Product and factor market competition limit the degree of economic concentration in regions with natural advantages. Continued concentration in many developed areas, and development of core areas without natural advantages, require other explanations such as the New Economic Geography (NEG) (Krugman 1991, 1993).

NEG models incorporate imperfect competition, increasing returns to scale, and mobile factors, making the concentration of economic activity endogenous. Labour resides in the region of employment and migrants respond mostly to labour demand. Continued agglomeration is facilitated by the preference for product variety among labourers. Yet, because it does not fully consider the diversity of factors underlying household location such as commuting or access to urban amenities, NEG may be limited in explaining 21st century urban hierarchy and core–periphery dynamics (Glaeser and Kohlhase 2004). Thus, we also consider competing theories for agglomeration of economic activity and the related spatial interactions.

2.1 *New Economic Geography population dynamics*

In NEG models, with their focus on pecuniary effects, close proximity to suppliers of intermediate inputs and customers lowers firm transportation costs (Venables 1996). Combined with

scale economies in the production of non-traded intermediate inputs (Fujita 1988), agglomeration economies arise. Yet, increased competition associated with proximity of economic activity acts as a dispersal force (Combes 2000), limiting agglomeration. Firms in the areas closest to agglomeration centres find themselves in a 'growth shadow' (Dobkins and Ioannides 2001), in which they are competitive in producing only those goods and services where they are not in direct competition with nearby higher tiered centres. Small cities serve their local markets, while larger cities serve wider markets that include small cities, leading to an urban hierarchy (Fujita and Thisse 1996).

NEG models predict that population in the distant hinterlands expands to provide the urban centre with farm products, until densities are sufficient for a new urban centre to emerge (Fujita et al. 1999). In a *mature*, more established urban system, such as the American urban system where the basic structure was defined circa 1920, the hinterlands would be approximately uniformly settled (as in CPT), with growth shadow effects as the key NEG mechanism for a positive distance–population *growth* relationship. If agglomeration forces in large urban centres dominate other growth factors in nearby areas, growth of the latter may be inhibited.

Modern agglomeration patterns reflect improved transportation and communication technology, shifts in trade patterns, and industry structural change. Industrial restructuring away from goods and towards services could produce vastly different patterns of spatial interactions compared with those based on 19th and early 20th century parameters (Glaeser and Kohlhase 2004). For example, falling transportation costs can have an ambiguous effect on agglomeration (Takatsuka and Zeng forthcoming). To the extent that agricultural goods become cheaper to ship into core areas, the tendency towards agglomeration increases (Fujita and Thisse 1996). Yet falling transportation costs can also disperse manufacturing activity to less congested areas (Desmet and Fafchamps 2005). Shifting international trade patterns can cause economic activity to move away from historic urban centres to border areas to reduce associated transportation costs (Hanson 1997). However, empirical evidence suggests a strong lock-in effect at the top of the urban system, with more churning at lower levels (Black and Henderson 2003).²

2.2 *Moving beyond the NEG framework*

The theoretical basis for considering factors beyond those related to NEG derive from the application of the compensating differential approach within a growth context. The compensating differential approach posits that households and firms respond to spatial differentials in consumptive and productive amenities, in which perfect mobility equalizes household utility of residence and firm profitability of production in spatial equilibrium (Roback 1982; Blomquist et al. 1988; Partridge and Rickman 2003). These other location specific factors can offset NEG shadow effects on growth to produce a positive relationship between population growth and *close* proximity to the largest agglomeration centres.

In their city model, Lucas and Rossi-Hansberg (2002) postulate that firms weigh the benefits derived from spatially concentrating to achieve closer input–output linkages against the gains of dispersing into residential areas to reduce commuting costs for their workforce. Extending this model to allow commuting and input–output externalities beyond the city captures more extensive spatial interactions (in which commuting costs rise and externalities decay over distance). Within this spatial contour, population growth is *inversely* related to distance from the urban centre rather than *directly* related as predicted by NEG growth shadows. Besides transport costs,

² Some NEG theories suggest catastrophic shocks could generate realignments of the entire urban system (Fujita and Mori 1997). However, Davis and Weinstein (2002) show that even the ruinous Allied bombing campaigns against Japan during the Second World War did not significantly alter its urban system. For a broader discussion of classical and neoclassical location patterns, see McCann (2002).

transactions costs such as information costs, about demand and locating trusted suppliers, also increase with distance from the core.³ Congestion costs, higher crime, pollution, and land prices (Glaeser 1997) likewise may push growth to nearby areas. Allowing for interregional migration would further extend the spatial reach of agglomeration.⁴

There also may be desirable urban amenities in higher-tiered urban areas such as greater diversity in consumption opportunities (Glaeser et al. 2001). The cost of accessing these amenities outside the region increases in terms of greater distances. Thus, an increase in amenities in higher-tiered urban areas or an increase in their valuation will make proximate regions more attractive to households, increasing net in-migration and population growth, while more remote regions may experience net out-migration and population decline (Graves and Mueser 1993; Mueser and Graves 1995).

2.3 *Synthesis of NEG and extensions for spatial population dynamics*

The predicted population dynamics of NEG models are altered by these extensions. Though subtly different than traditional CPT, many NEG models predict a hierarchy of cities, in which the availability of services increases when moving towards the top of the hierarchy (e.g., Fujita et al. 1999). This occurs because of rising demand thresholds for higher-order goods and services and spatial competition among firms, the latter acting as a nearby dispersal force. Each tier of city has services that are available in lower-tier cities plus those for which threshold requirements are first met at that tier. The top tier (n) in the hierarchy offers the full range of services including the highest-order financial and legal services, with the first or lowest tier offering only the most basic such as petrol. Residents and businesses in the first tier travel to the nearest higher-tier cities for second, third, and higher orders of services.

In NEG models, this hierarchy generates agglomeration growth shadows, where spatial competition near higher-tiered centres constrains the growth of local businesses.⁵ Yet, because of the extensions discussed above, spatial-competition shadows may be overcome by offsetting positive agglomeration spillovers. Rural areas and smaller urban centres can benefit from close proximity to successively higher-tiered centres, while incurring penalties for increasing remoteness, having to travel incrementally farther to access increasingly higher-orders of services and to access employment through commuting. While they may not compete with the higher tier centres for the local recruitment of the highest order goods and services, a location that affords them ready access will have advantages.⁶

The distance penalties/benefits for community i in tier j of n tiers of cities in the urban hierarchy evolve as follows. Each consecutive higher-tier of urban areas ($j + 1, j + 2 \dots n$) has successively higher orders of services and urban amenities. Consistent with marginal costs, the incremental distance is defined as the difference in the distances between places in the tiers j and $j + 1$; $j + 1$ and $j + 2$; $j + 2$ and $j + 3$; and so on. For each i , beyond the 'own' level j of urban centre, let the incremental distance to the nearest place in the higher-tier equal d^i and the marginal 'penalty' for population growth of greater incremental distance from the previous tier

³ Wolf (1997) finds that US domestic goods are shipped an average of 255 miles.

⁴ Housing costs have also been added to NEG models, increasing the cost of living in agglomerated areas and acting as a centrifugal force limiting their size (Helpman 1998).

⁵ In contrast to central place theory, because of its use of the iceberg transport cost assumption, NEG models predict an area's sales to other locations diminish with greater distance. Spatial competition in NEG can lead to the need for product differentiation, which can produce a hierarchy of cities where the largest cities specialize in the higher order goods and services where they have a cost advantage.

⁶ Some of these effects could vary spatially and across regions. For an example of a geographically weighted regression approach that partially considered these issues, see Partridge et al. (2007, 2008a).

equal to ϕ^j .⁷ The different ϕ^j across tiers allows the total distance response to have different segments (not just one linear response for the entire hierarchy). The sum (over all tiers) of the accessibility penalties for population growth at location i in the j th tier can be depicted as:

$$Penalty_{ij} = \sum_t d^t \phi^t, \quad (1)$$

where the summation is over $t = j + 1$ to the n^{th} tier. The penalty term reflects the total cost a region faces due to cumulative incremental distances as a result of travelling up the hierarchy to access progressively higher-order services.

3 Empirical implementation

The underlying agglomeration mechanisms discussed above vary in their predictions regarding the effect of distances from agglomerated core areas on performance of peripheral areas. Our primary interest is the impact of geographic position relative to nearby cities (differentiated by hierarchical level) on local growth processes.

We regress population growth between periods 0 and t on initial period (0) conditions to allow for transitions from one equilibrium to another, consistent with related literature (Eeckhout 2004; Partridge et al. 2008b).⁸ Specifying subsequent population growth as a function of the (pre-determined) initial-period characteristics mitigates concerns with statistical endogeneity. Regarding the key distance variables, we assume that the US urban hierarchy was well established by 1990. This assumption follows Tabuchi and Thisse (2006) in a theoretical NEG framework, as well as empirical evidence that suggests that the urban hierarchy has been very stable over time (e.g., Eeckhout 2004; Duranton 2007). Thus, distances from a location to its higher-tiered urban areas are predetermined and set by prior development of the US urban system. Yet, statistical endogeneity in other variables may bias the distance results, a topic we consider in more detail below.

Population change is examined primarily over the 1990–2000 period, in which sensitivity analysis is conducted for 1990–2006. Counties in the lower 48 states and the District of Columbia are our units of observation.⁹ We use multiple sub-samples to examine the different transmission mechanisms across the urban hierarchy. Counties occupying the lowest tier in the urban hierarchy are those that are not part of either a micropolitan area (MICRO) or a larger metropolitan area (MA). MICROS, small ‘urban’ centres of 10,000–50,000 people and counties with tight commuting linkages to them, are considered the next tier.¹⁰ Moving up the urban hierarchy, we next divide the sample into counties in MAs of less than and more than 250,000 people. NEG theory posits that small and medium-sized urban centres need distance protection

⁷ For the nearest place in higher tier $j + 1$, the incremental distance equals the *actual* distance between area i and the nearest area in tier $j + 1$. Section 3 provides specific examples. If there are growth shadows, the ‘penalty’ could revert to a benefit.

⁸ Examining growth rates has the advantage of differencing out fixed effects associated with levels or the scale of the locality (Hanson 2001).

⁹ Following the US BEA, there are cases where independent cities are merged with the surrounding county to form a more functional economic area (especially in Virginia). Forty-three mostly small rural counties are omitted due to the lack of economic data. For details, see Partridge and Rickman (2006).

¹⁰ We generally use the 2003 MA/MICRO definitions, as MICROS were first defined in 2003. An inclusive definition of MAs is desired to isolate growth due to changing commuting patterns versus *intra* urban centre interactions due to other factors. Note, *excluding* the fastest growing recently-acquired outer MA counties from the rural sample, *weakens* any underlying negative rural-distance to urban centre response (actually strengthening our results). Sensitivity analyses use earlier 1999 definitions mostly based on 1980s commuting patterns to establish boundaries for the existing (1990) MAs, along with MAs newly defined in the 1990s.

from the higher-order centres to reduce spatial competition (Fujita et al. 1999). The 250,000 population threshold (we conduct sensitivity analysis using different thresholds) splits the metro sample into two approximately equal groups.

The most complete reduced-form specification for county i , located in state s is depicted as:

$$\begin{aligned} \% \Delta \text{POP}_{is(t-0)} = & \alpha + \delta \text{POPDEN}_{is0} + \phi \text{GEOG}_{is0} + \theta \text{DEMOG}_{is0} + \psi \text{ECON}_{is0} \\ & + \gamma \text{AMENITY}_{is0} + \sigma_s + \varepsilon_{is(t-0)}, \end{aligned} \quad (2)$$

where *POPDEN* is initial-period population density to control for own-county agglomeration or congestion effects. *GEOG*, *DEMOG*, *ECON*, and *AMENITY* are vectors that represent: geographic attributes including distance to different tiers in the urban hierarchy; demographic characteristics; economic characteristics; and amenities. The regression coefficients are α , δ , ϕ , θ , ψ , and γ ; σ_s are state fixed effects that account for common factors within a state; and ε is the residual.

Our analysis begins with more parsimonious models than Equation (2) to assess whether potential multicollinearity and endogeneity affect the key results. The most parsimonious models include only variables that are clearly exogenous (e.g., climate) or predetermined (distance), while additional factors are added in successive models to assess robustness. The county residual is assumed to be spatially correlated with residuals for neighbouring counties, with the strength of the correlation being inversely related to the distance between the two counties. We use a generalized method of moments (GMM) procedure to produce t -statistics that are robust to cross-sectional spillovers (Conley 1999).¹¹ Appendix Table 1 presents detailed variable definitions, sources, and descriptive statistics.

GEOG contains several measures of proximity to higher-tiered urban areas. The first measure is distance to the nearest urban centre of any size, which can be either a MA or MICRO. If the county is already part of an urban area, this is the distance from the population-weighted centre of the county to that of the urban area – for example, for a suburban county of a large MA, this would be the distance to the core of the urban area and would reflect various sprawl or commuting patterns.¹² Because these are *intra*-urban effects, these *within* MA distance effects are not germane to our discussion of distance effects or growth shadows. Conversely, if the county is not part of an MA or MICRO, it is the distance from its centre to that of the nearest urban place. Outer counties *within* MAs or MICROs areas could grow faster due to urban sprawl and suburbanization. Yet, greater remoteness to the next *higher* urban tier may promote or detract from growth depending on whether distance is costly or offers protection from spatial competition.

To reflect ‘penalties’ for remoteness from additional tiers, we include incremental distances to higher-tiered urban centres for counties whose nearest city is not the highest tier. First, we include the incremental distance from the county to reach an MA.¹³ Next we include variables that measure the incremental distance to reach an urban centre of at least 250,000, at least 500,000, and >1.5 million people.¹⁴ The incremental distances reflect the additional penalty (or benefit) a resident or business of a county encounters because they incur costs to access offerings

¹¹ The bandwidth extends 200 kms, after which zero correlation in county residuals is assumed.

¹² The population weighted centroid of each county is from the US Census Bureau. The population category for MAs is based on initial 1990 population. If the urban centre only has one county, this distance is zero.

¹³ For example, if rural county A is 40 kms from a MICRO and is 70 kms from the nearest MA, the incremental distance to the nearest MA would be 30 kms. Conversely assume county B is located in a MICRO, being 25 kms from the centre of its MICRO and 70 kms from the nearest MA. The corresponding incremental value to the nearest MA would be 45 kms (70 – 25). For a MA county, the incremental value is zero.

¹⁴ Incremental distance is calculated as before. If the county is already nearest to a MA that is in either a larger or same size classification, then the incremental value is zero. For example, if the county’s nearest urban centre of any size (or MA of any size) is already over 500,000, then the incremental values for the at least 250,000 and at least 500,000 categories are both equal to zero.

in progressively higher-ordered urban centres. The largest urban tier represents top-tier regional/national centres, while the other smaller-centre sizes capture different-size labour markets (for commuting) and access to personal and business services.

In some MICRO and MA samples, we include distance to the nearest urban centre *within* the same tier. For a given MICRO, this would be the distance to the nearest other MICRO. The sign of this coefficient shows the net effect of two offsetting possibilities: (1) spatial competition among urban centres *within the same tier* (akin to a growth shadow), or (2) close proximity to another urban centre in the same tier enhancing the regional agglomeration effect. For large MAs, the own-tier distances are calculated for population categories of 250,000–500,000, 500,000–1.5 million, and >1.5 million.

The sum of the (marginal) distances equals the distance to a MA with at least 1.5 million people. The distance variables in segments are the cost of reaching the nearest *highest*-tiered city. The individual elements of equation (1) show that the *impact* of distance is decomposed into segments to each intervening tier city, potentially producing a non-linear relationship, namely, distance effects vary depending on the urban tier is being considered.

Figure 1 illustrates the distance measurements for Park County, a rural Montana county containing about 15,000 residents in 1990. The nearest urban area is Gallatin County (a MICRO), containing the city of Bozeman, located 42 kms away. The nearest MA is Billings, located 158 kms away, or an incremental distance of 116 kms (158 – 42). Because Billings is a

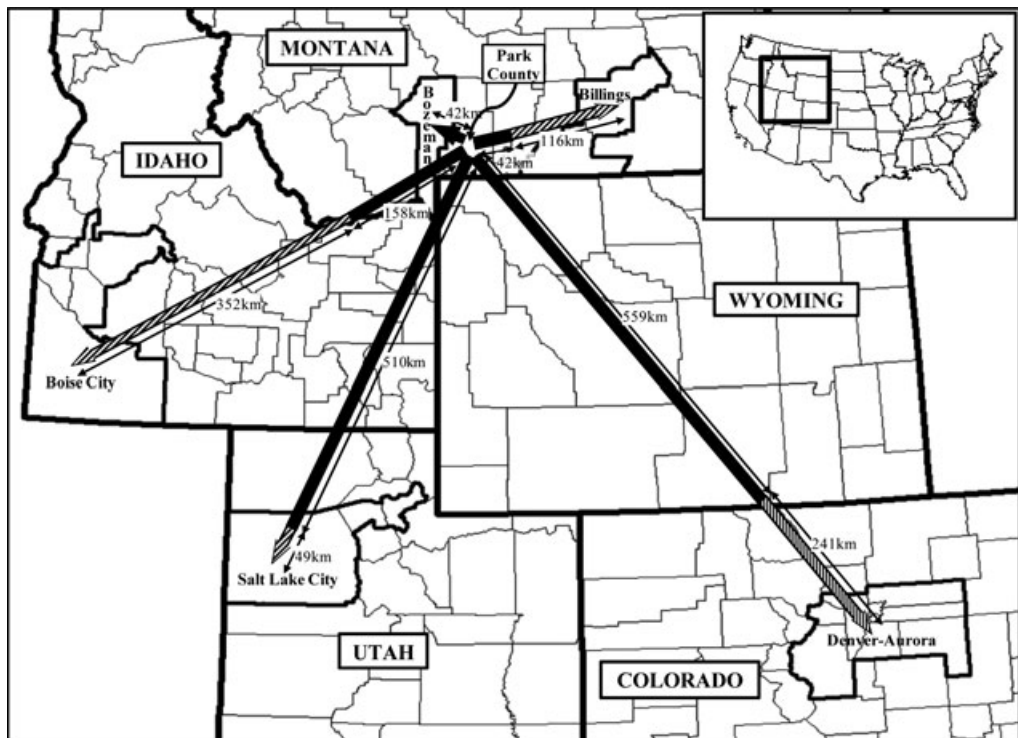


Fig. 1. Calculation of the distance measures for Park County Montana

Notes: This figure shows the incremental distance penalties from Park County, Montana, a sparsely populated county at the lowest tier of the hierarchy. Bozeman, Montana is the nearest urban area (a micropolitan area) that is 42kms away. Using 1990 populations, the subsequently higher-ordered tiers are Billings, MT (an MA < 250,000 population); Boise, Idaho (MA population between 250,000–500,000); Salt Lake City, Utah (MA population between 500,000–1.5 million); and Denver, Colorado with over 1.5 million population. After Bozeman, the grey hatched segments show the incremental distance variables used in the empirical model.

small MA, with about a 1990 population of 121,500, Park County residents and businesses are also affected by agglomeration economies in the nearest larger MA of at least 250 K, which is Boise, Idaho, with 1990 population of 319,500. Boise is 510 kms from Park County, but the incremental distance to agglomeration economies not found in Billings is 352 kms (510 – 158). The nearest MA of at least 500 K, the next higher tier above Boise, is Salt Lake City, Utah, with roughly 768,000 people in 1990. Salt Lake is 559 kms away from Park County, an incremental distance of 49 kms (559 – 510). Denver, Colorado is the nearest highest tier city with over 1.5 million (1990 population of 1.65 million). Denver is 800 kms away from Park County, or an incremental distance of 241 kms (800 – 559).

Other variables in the **GEOG** vector include population of the nearest (if a rural county) or own (if **MICRO** or MA county) urban centre. A county may benefit from proximity to a larger nearby urban centre if more positive agglomeration effects spill over (labour market effects for commuting and proximity to amenities and higher-order services). The existence of growth shadows would produce offsetting responses.

Analogous to the distance variables, we also include incremental population variables for the nearest MA, a MA of at least 250,000, at least 500,000 and at least 1.5 million, people.¹⁵ Because we already include the incremental *distance*, these population terms account for any *marginal* population impact. That is, they account for *within* tier effects of urban size, while the incremental distance terms account for the penalties of reaching an urban centre of at least the specified size (threshold effects).¹⁶ Other specifications (below) use the actual rather than incremental population, as well as models that omit incremental population altogether, but there was almost no change in the key incremental distance results. Finally, some models control for the population in surrounding counties within the county's BEA economic region (see footnote a, Appendix Table 1) to account for factors such as agglomeration spillovers and market potential (Head and Mayer 2004, 2006).

The remaining control variables capture potential causes of population change aside from geographic location. First, we account for natural amenities (**AMENITY**) as measured by climate, topography, percent water area, and a related amenity scale constructed by the US Department of Agriculture (see Appendix Table 1). Amenities are included in all models as they reflect natural location advantages.

To examine robustness, we also include numerous demographic and economic variables (in 1990) in some models. To account for human capital migration effects, we include initial-period **DEMOG** measures of racial composition, past immigration, age, and educational attainment. To control for disequilibrium economic migration, some models incorporate the following **ECON** measures: 1989 median household income, 1990 unemployment rate, 1990 employment shares in agriculture and in goods production. We also include the 1990–2000 industry mix job growth, a common exogenous measure of demand shifts.¹⁷ To account for nearby county economic spillovers, some models include BEA-region values of median income, unemployment, and industry-mix growth measures (excluding the county of interest).

¹⁵ For example, if the nearest/actual urban centre is 45,000 (**MICRO**), the next closest urban centre is 600,000, the third closest urban centre is 2 million people, then the incremental population of nearest MA is 555,000 the incremental population of a MA that is >250,000 is 0, the incremental population of a MA >500,000 is 0, and the incremental population of a MA that is at least 1.5 million is 1.4 million (2 – 0.6 million).

¹⁶ For example, for the 250,000 cutoff, the incremental distance to an urban centre variable accounts for penalties to reach an urban centre of at least 250,000. The incremental population variable accounts for any marginal spillovers due to this urban centre/tier having a population in *excess* of 250,000.

¹⁷ Industry mix employment growth, derived from the shift-share model, is the sum of the county's initial industry employment shares multiplied by the corresponding national industry growth rates for the period. Because national industry growth and initial employment levels should be exogenous to county industry growth, it is a common instrument for local job growth (Bartik 1991; Blanchard and Katz 1992; Bound and Holzer 2000).

In other models, state fixed effects are included to account for factors such as policy differences, geographic location with respect to coasts, and settlement period.¹⁸ When state fixed effects are included, the other regression coefficients are interpreted as the responses to *within* state differences in the explanatory variables.

4 Empirical results

Descriptive statistics are reported in Appendix Table 1, whereas Appendix Table 2 has selected sub-sample statistics. Mean county population growth rates (Appendix Table 2) are the lowest for rural counties, followed by micropolitan areas, small and large MAs. The large standard deviations reveal there is a great deal of variability within groups, and it is this variability that is of primary interest in the analysis that follows. The mean distance to the nearest or actual MICRO or MA of 35 kms (Appendix Table 1) is indicative of the mean proximity of the counties in our study to some urban centre, while the set of mean *incremental* distances to successively higher tier places show that the full urban hierarchy is much more remote, additively 258 kms. This total mean distance as well as the mean distance to 'own tier' counties of 43 kms, will be a weighted average of the 1,300 noncore rural counties, 672 micropolitan counties, 416 that are part of MAs of less than 250,000 population and the 641 that are greater than 250,000. Table 1 contains the econometric results for the 1990–2000 period for counties located in noncore rural areas, MICRO areas, small MAs with less than 250,000 people, and large MAs with population over 250,000. Sensitivity analysis follows thereafter.

4.1 Base regression models

With the exception of the large MA group, we start with a very parsimonious model that only includes distance and amenity measures, followed by a second model that adds the urban population, economic, and demographic variables, and a third model that adds state fixed effects. These models successively test the robustness of the results to alternative causes of population change such as initial economic conditions, or econometric concerns regarding omitted variables, multicollinearity, endogeneity, or omitted autoregressive processes. Stability of our key results across these varied specifications will show that our findings are not an artifact of a particular econometric (mis)specification.

In Table 1, the key distance results across columns (1)–(3) for rural counties, (4)–(6) for MICRO counties, and (7)–(9) for small MA counties are very similar, indicating the robustness of the results.¹⁹

Given their robustness, we describe the results of the fully specified models in columns (3), (6), (9), and (10). We start with discussion of the rural areas and then move to discussing the smaller urban centres and the largest MAs. We focus our discussion on the distance results due to their first-order importance.

¹⁸ We experimented with also including three indicators for being within 50 kms to the Atlantic Ocean, Pacific Ocean, and the Great Lakes. However, our results were essentially unaffected and the ocean/lake proximity variables were generally insignificant suggesting that state fixed effects adequately represent these natural advantage effects. Other effects such as access to air and rail transportation are part of those urban hierarchy effects we wish to capture with our distance measures.

¹⁹ The large MA sample results are also robust, but we do not report all of them for brevity.

Table 1. Dependent variable: percentage change in US county population 1990–2000

Variables/var groups	Noncore rural area			Metropolitan area			Small MA with pop ≤250,000			MA >250,000
	Dist	Add other X	Add state FE	Dist	Add other X	Add state FE	Dist	Add other X	Add state FE	Full model
Intercept	-0.508 (-0.10)	17.344 (0.87)	19.482 (1.09)	2.200 (0.38)	29.742 (1.04)	45.870* (1.82)	-3.489 (-0.43)	46.031 (1.43)	11.409 (0.39)	55.262** (2.09)
Distance to nearest or actual urban centre	-0.075** (-4.32)	-0.092** (-5.98)	-0.102** (-7.62)	0.054 (0.97)	0.216** (3.31)	0.137** (2.14)	0.092** (2.17)	0.082 (1.53)	0.064 (1.55)	0.188** (4.26)
Inc Dist to MA	-0.050** (-4.87)	-0.040** (-4.51)	-0.038** (-4.79)	-0.039** (-2.99)	-0.037** (-3.25)	-0.026** (-2.02)	n.a.	n.a.	n.a.	n.a.
Inc Dist to MA > 250 k	-0.026** (-4.91)	-0.025** (-5.67)	-0.030** (-5.17)	-0.028** (-3.43)	-0.036** (-4.95)	-0.029** (-3.24)	-0.028** (-2.95)	-0.038** (-4.17)	-0.057** (-5.53)	n.a.
Inc Dist to MA > 500 k	-0.006 (-0.71)	-0.004 (-0.50)	-0.018** (-2.42)	-0.003 (-0.25)	-0.008 (-0.75)	-0.009 (-0.78)	0.004 (0.24)	0.002 (0.18)	-0.027** (-2.23)	0.011 (-3.3E-04)
Inc Dist to MA > 1,500 k	-0.006 (-1.22)	-0.007 (-1.32)	-0.011** (-2.50)	-0.000 (0.00)	0.004 (0.73)	0.003 (0.51)	-0.001 (-0.09)	-0.007 (-0.92)	-0.026** (-2.66)	-0.004 (-0.04)
Dist to nearest own tier	0.021 (0.75)	-0.019 (-0.83)	0.003 (0.15)	0.026 (0.87)	0.026 (1.30)	0.015 (0.85)	0.020 (0.99)	0.045** (2.69)	0.044** (3.14)	0.013 (0.74)
Population density	N	-0.066** (-2.19)	-0.054* (-1.95)	N	-0.013 (-1.06)	-0.023* (-1.81)	N	-0.018** (-3.17)	-0.022** (-4.07)	4.2E-04* (1.74)
Pop of nearest or actual urban centre	N	9.3E-07 (0.18)	6.8E-06 (1.49)	N	-5.8E-05** (-2.75)	-2.3E-05 (-1.18)	N	1.6E-05 (1.07)	4.6E-05** (3.45)	-5.0E-07 (-1.50)
Inc pop of nearest MA	N	-1.7E-06* (-1.86)	-1.4E-07 (-0.18)	N	-1.4E-06 (-1.47)	-9.6E-07 (-1.14)	N	n.a.	n.a.	n.a.
Inc pop of MA > 250 k	N	-1.2E-06* (-1.94)	-5.5E-07 (-0.95)	N	-6.4E-07 (-0.93)	-4.9E-07 (-0.76)	N	6.2E-07 (1.08)	8.9E-07* (1.79)	n.a.
Inc pop of MA > 500 k	N	-1.1E-06** (-2.00)	-4.6E-07 (-0.96)	N	-4.3E-07 (-1.13)	-2.6E-07 (-0.66)	N	-1.0E-06** (-2.25)	-2.1E-07 (-0.50)	-3.1E-07 (-0.98)
Inc pop of MA > 1,500 k	N	-6.9E-07** (-3.62)	-2.3E-07 (-1.07)	N	-6.2E-07** (-4.03)	-2.3E-07 (-1.44)	N	-8.8E-07** (-2.39)	2.2E-07 (0.57)	-5.0E-07** (-2.69)
Pop in surrounding counties	N	2.4E-07 (0.77)	-2.0E-07 (-0.74)	N	4.1E-08 (0.08)	2.5E-07 (0.52)	N	-7.6E-08 (-0.19)	-4.5E-07 (-1.34)	4.0E-07* (1.93)
Weather/Amenity ^a	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Economic/Demographic ^b	N	Y	Y	N	Y	Y	N	Y	Y	Y
Surrounding Econ/Demog ^c	N	Y	Y	N	Y	Y	N	Y	Y	Y
State fixed effects (FE)	N	N	Y	N	N	Y	N	N	Y	Y
R ²	0.33	0.49	0.60	0.24	0.52	0.63	0.17	0.51	0.64	0.63
No. of counties	1,300	1,300	1,300	672	672	672	416	416	416	641
F-statistic										
All MA pop = 0	N	3.10**	2.00*	N	3.99**	0.53	N	2.87**	2.79**	1.15
Inc MA pop = 0	N	3.70**	0.53	N	2.69**	0.39	N	3.69**	1.03	1.66
Inc distance to MA = 0	27.88**	17.19**	12.27**	8.90**	12.23**	4.33**	4.73**	6.04**	7.53**	0.36

Notes: Robust t-statistics from Conley (1999) estimator are in the parentheses. A ** or * indicates significance at ≤5% or ≤10% level respectively. N = not included, Y = included. ^a sunshine hours, January temp, July humidity, typography, amenity ranking, and percent water area.

^b 1989 median household income, 1990–2000 industry mix emp. growth, 1990 unemp. rate, 1990 share ag. emp., 1990 share goods emp., 6 age-distribution variables for 1990, 4 education categories for 1990, 5 race/ethnicity variables for 1990, and percentage of population immigrated during 1985–90.

^c weighted average 1989 median household income, 1990–2000 industry mix emp. growth, and 1990 unemp. rate in surrounding counties within a BEA region.

4.1.1 Urban–hinterland interactions

The distance to the nearest urban centre coefficient (column 3) suggests that for a rural county, every kilometre increase in distance from its nearest urban centre (of any size) is associated with 0.1% less population growth. When measured at the mean distance of 60 kilometres (Appendix Table 2), this translates into about 6% less population growth than a rural county adjacent to the urban centre's core. The influence of the urban hierarchy does not end there. If the nearest urban centre is only a MICRO area, the rural county loses an additional 0.038% of population growth per incremental kilometre to reach an MA of any size. If this urban centre is a small MA of less than 250,000 people, there is another penalty of 0.03% per incremental kilometre to reach a MA of at least 250,000. There are corresponding population growth penalties of about 0.02% and 0.01% per kilometre to reach MAs of at least 500,000 and 1.5 million people. At the mean incremental distances (Appendix Table 2), the typical rural county incurs a distance penalty of 11.9% in population growth during the 1990s for its remoteness from the combined urban tiers.^{20, 21}

Although it is notable that there are incremental penalties for a hinterland county being remote from urban centres up to 1.5 million, the largest marginal penalty is for remoteness from *any* urban centre with population of 10,000 or more. This suggests that accessibility for commuting and basic urban services may be of over-riding importance. This is inconsistent with an urban growth shadow or with a long-run CPT equilibrium with uniform population growth in the hinterlands. Such US settlement patterns illustrate the strong far-reaching regional forces that can encourage sprawl into rural areas located near urban areas.

The results for the 1990–2006 period are similar, generally being larger in proportion to the years added to the analysis (Appendix Table 3). Exceptions include the coefficient for population of the nearest urban centre becoming significantly positive and the coefficient for population density becoming insignificant.

4.1.2 Small urban centre spatial interactions

The MICRO and small MA results in columns (6) and (9) reveal that more distant counties *within* an urban centre are growing faster, consistent with sprawl and suburbanization (though the small MA result is only marginally significant). This pattern applies even after accounting for a host of other characteristics. For MICROs, the incremental distance to the nearest MA coefficient implies a 0.026% population penalty per every kilometre farther away. Beyond that, only the incremental distance to an urban centre greater than 250,000 has a statistically significant negative impact.

For small MAs, the estimates suggest that for increased incremental distance from urban centres greater than 250,000, 500,000, and 1.5 million people, small MAs are marginally penalized even more than rural counties. For example, at the mean incremental distances (Appendix Table 2), small MAs *ceteris paribus* incur growth penalties of 8.3%. This penalty likely relates to a lack of access to higher order services and amenities.

²⁰ The results are derived by taking the coefficients from Table 1 multiplied by the means from Appendix Table 2 and summing these figures to obtain a distance effect. For the rural distance penalty, when using all coefficients, the results correspond to taking the sum of $-0.102(59.91) - 0.038(43.47) - 0.030(76.02) - 0.018(45.32) - 0.011(83.45)$, which equals -11.9 .

²¹ A one standard deviation increase in the distance to the nearest urban centre is associated with an expected 3.1% decline in rural county population growth, while a one-standard deviation increase in all of the incremental distance terms is associated with 10.5% less growth.

The sole evidence of a distance-based growth shadow is for small MAs being too close to other small MAs *within* their tier. So, greater distance of a small MA from other small MAs allows it to grow more. For the mean distance between small MAs, population growth is 2% slower, in which a one standard deviation increase in the distance increases growth by 2%.

The urban-centre *population* results are more ambiguous. For both MICROs and small MAs, county population growth is inversely related to own initial population density. Inconsistent with growth shadows, the incremental population variables are mostly insignificant. In fact, for small MAs, the incremental population of the nearest MA greater than 250,000 positively and significantly affects population growth. Yet, neighbouring region population is insignificant in the MICRO and small MA models.

Expanding the sample to 2006 generally produced comparable results (Appendix Table 3). For micropolitan areas, population density became insignificant. The urban hierarchy coefficients for small metropolitan areas became more negative by magnitudes which are greater than the proportion of the sample period expansion, and much greater than what occurred for noncore rural areas and micropolitan areas. Thus, not only were small metropolitan areas penalized more for greater distances from high-tiered metropolitan areas, the relative penalty appears to be increasing in the most recent years.

4.1.3 Large urban centre spatial interactions

Column (10) of Table 1 shows the corresponding results for the counties located in large MAs (population >250,000). With the exception of the positive *within* MA distance coefficient, none of the proximity variables are significant. Thus, the general pattern is one of sprawl within a given metropolitan area, while proximity to higher-tiered MAs and other same-tiered MAs plays little role. The sprawling pattern suggests that two nearby metropolitan areas could grow/sprawl into one larger metropolitan area if they are sufficiently close – for example, Baltimore–Washington DC – though this sprawl pattern is a very different spatial interaction than the process predicted by NEG.

The positive own-county population density variable suggests that there are favourable local agglomeration effects at the 10% level, consistent with a size threshold where own agglomeration effects become more important for larger urban centres. The positive and significant (at the 10% level) neighbouring-county population coefficient suggests that agglomeration benefits are also derived regionally. There is now evidence of a growth-shadow effect for counties located in MAs of between 250,000 and 1.5 million people when they are close to a MA with population over 1.5 million (evidenced by the negative effect of increased population of the large MA).²² Given the corresponding insignificant distance effects, it may be that only the largest MAs in the top tier cast a growth shadow consistent with NEG. The insignificant incremental distance findings still are most consistent with no clear spatial interactions between neighbouring cities (Ioannides and Overman 2004).²³

The results for the expanded period of 1990–2006 suggest that urban sprawl appears to be pervasive. Population density also has a more significant positive effect on growth, while population in surrounding counties is now insignificant. Yet, the general pattern is very similar to the shorter 1990–2000 period.

²² Dividing small and large MAs at 500,000 was considered in sensitivity analysis, but without much affect.

²³ We also considered whether using earlier 1993 and 1999 MA boundaries to define our sample (MICRO areas were not defined in 1999) would affect our key distance results (not shown). These boundaries would have included only commuting patterns from the 1980s identified in the 1990 Census, while the 1999 boundaries include any new MAs defined in the 1990s. In defining these small and large MA samples, either the 1993 or 1999 boundaries are used in determining the MA's particular population category (e.g., less than or greater than 250,000). Nonetheless, the distance results were generally robust using these boundaries with only modest changes.

4.1.4 Further search for growth shadows

To further search for the existence of growth shadows, we added interactions of the distance variables with the corresponding urban centre population (not shown). This tests whether growth shadow effects attenuate with distance. These interaction variables were jointly insignificant in the MICRO and small MA models, but significant at the 5% level in the rural and large MA cases.

For rural counties, the negative distance–population interaction coefficients generally suggest less population growth the more distant the county is from an urban centre of a given population level, which is the opposite of an agglomeration growth shadow. For the ‘smaller’ large MAs with between 250,000 and 1.5 million people, the positive interaction coefficient suggests that greater distance from an urban centre of >1.5 million mitigates the adverse population effects described above, consistent with the growth shadow interpretation above.

4.2 Summary of findings

We generally find that population growth for counties in rural areas, MICROs, and small MAs is positively related to proximity to a higher–tier urban centre. This is inconsistent with growth shadows of CPT and NEG models. There is less evidence that (regional) market potential is a good explanation for population change because the own population density and the neighbouring county population coefficients have the wrong sign or are insignificant. One reason for the mixed pattern could be ‘exports’ from smaller urban centres and the hinterlands are more directed to national and international markets than those nearby.

For larger MAs, we find surprisingly little evidence that proximity to (even) higher–tier urban centres affects their growth. The pattern is consistent with offsetting effects from decaying agglomeration effects and CPT factors. There is evidence of growth shadow effects between small MAs *within* the same tier. There also is some evidence that among ‘medium’ MAs of 250,000 to 1.5 million, there are adverse competition effects (growth shadows) from proximity to the very largest urban centres.

Examination of 1990–2006 confirms the continued negative effects of a county’s distance from higher–tiered urban areas. In fact, the effects for small MAs became more–than–proportionately negative (relative to the length of the time period expansion) – namely, they appear to be intensifying after 2000. Sprawl also appears to be continuing at the outer fringe of MAs with greater than 250,000 people. However, generally the patterns are similar over the longer period.

4.3 Commuting linkages

The lack of urban growth shadows could relate to tight commuting links between cities, rather than geographically–widespread input–output linkages.²⁴ Most agglomeration models suggest that urban input/output externalities are a function of thresholds and the size of the urban centre (Black and Henderson 2003). Marginal growth in an urban centre would not measurably affect the size of its input–output externalities – for example, New York MA’s input–output

²⁴ For example, past research suggests that knowledge spillovers have a much smaller geographic scope, perhaps less than a few miles because of their highly personalized nature (Rosenthal and Strange 2001, 2003).

externalities would be only marginally affected if it grew by (say) 5%. Yet, if it created 5% more jobs, this may create commuting opportunities in nearby counties, increasing their population growth.

By controlling for job growth in the urban centre, we can ascertain how much of its job growth spills over and creates opportunities in neighbouring counties. Thus, if the incremental (urban centre) distance coefficients are much smaller in magnitude when the urban commuting measures (employment growth) are included in the model, this would suggest that the incremental distance attenuation affects are mostly due to commuting. Any remaining distance effects would more likely be related to spillovers from threshold effects (input/output externalities and access to urban amenities).

To assess this issue, we include measures of urban centre job growth for the same urban–tier categories used above. Because commuting effects likely die out after 160 kilometres (100 miles), we set the corresponding nearest urban centre employment growth equal to zero if it is farther than 160 kilometres from the county. Even within 160 kilometres, commuting effects will decay with distance. Hence, we also include interactions of the nearest urban centre's job growth with the county's distance from it. Finally, population and job growth may be simultaneously determined. To account for this, we substitute the relevant 1990–2000 industry mix employment growth as an exogenous proxy for local job growth.

The results for the four size groupings are reported in Table 2. For each group, the first set of results is for the slightly-adjusted base model from Table 1, which is reported only for comparative purposes.²⁵ The second model for each group adds the industry mix growth rates for the nearest/actual urban centre in each size category and the corresponding industry mix–distance interactions.

In the noncore rural model, the industry mix terms for the nearest urban centre are consistently positive and jointly significant at the 5% level, suggesting nearby urban job growth creates rural commuting opportunities.²⁶ Likewise, the urban centre distance–industry mix interaction terms are all negative and jointly significant at the 5% level. As expected, positive urban employment growth effects attenuate with distance. Yet, the incremental distance to urban centre coefficients are still jointly significant at the 5% level, while their magnitude declines only modestly. The rural pattern suggests that urban job growth spreads out and lifts rural population growth through commuting opportunities. However, because the incremental distance coefficients remain large, there appear to be important backward–forward externalities that affect rural population growth.

Regarding the urban centres, the industry mix employment growth terms and corresponding distance interactions are almost universally insignificant. So job growth in nearby higher–tier urban centres generally has little statistical influence on smaller urban centre growth – suggesting that commuting ties do not underlie the interactions between urban centres. The pattern remains that growth in MICROS and small MAs is inversely related to distance from higher–order urban centres, consistent with a predominant role for accessibility to urban centres, one that dominates any potential shadow effects. For larger urban centres, the results still suggest relatively little spatial interaction. Since commuting does not appear to be a strong contributing factor, one of our conclusions is that there are geographically far–reaching backward–forward externalities between smaller and larger urban centres.²⁷

²⁵ With the exception of the noncore rural sample, the own–county industry mix measure is also omitted from these models as the urban centre's overall industry mix job growth accounts for localized employment growth.

²⁶ For the variables reported in the other tables, the coefficients are similar to the base model in Table 1, and are not reported for brevity, while the distance to the nearest tier variable is omitted.

²⁷ The insignificance of the industry mix employment growth variables for the urban samples supports our interpretation that they reflect commuting linkages and not static spatial input–output linkages, the latter expected to exist *between* urban areas rather than *between* urban and less–populated rural areas.

Table 2. Dependent variable: percentage change in US county population 1990–2000

Variables/var groups	Noncore rural		Metropolitan area		Small MA ≤ 250,000		Large MA > 250,000	
	Base model	Full model	Base model	Full model	Base model	Full model	Base model	Full model
Intercept	24.225 (1.32)	25.195 (1.39)	92.418** (3.66)	62.614** (2.79)	32.797 (0.95)	28.226 (0.85)	84.577** (2.93)	77.256** (2.64)
Dist to nearest or actual urban center	-0.103** (-7.96)	-0.070** (-3.50)	0.006 (0.09)	-0.144 (-0.56)	0.054 (1.33)	0.033 (0.63)	0.173** (3.92)	0.198** (4.15)
Inc Dist to MA	-0.039** (-5.27)	-0.015* (-1.67)	-0.034** (-2.61)	-0.043** (-2.39)	n.a.	n.a.	n.a.	n.a.
Inc Dist to MA > 250 k pop	-0.030** (-5.77)	-0.026** (-4.79)	-0.033** (-3.40)	-0.026** (-2.54)	-0.065** (-5.66)	-0.084** (-5.42)	n.a.	n.a.
Inc Dist to MA > 500 k pop	-0.019** (-2.68)	-0.013 (-1.63)	-0.009 (-0.69)	-0.017 (-1.22)	-0.024* (-1.70)	-0.034* (-1.90)	-0.005 (-0.42)	0.005 (0.22)
Inc Dist to MA > 1,500 k pop	-0.011** (-2.64)	-0.007 (-1.62)	0.006 (0.89)	-0.001 (-0.12)	-0.021* (-1.75)	-0.022 (-1.45)	-0.004 (-0.54)	-0.008 (-0.70)
Industry mix growth 1990–2000	98.669** (6.79)	95.338** (6.67)	N	N	N	N	N	N
Indmixgr of metropolitan area	N	4.675 (0.33)	N	152.214** (5.99)	N	N	N	N
Indmixgr of MA < 250 k	N	29.895** (2.70)	N	-20.727 (-1.29)	N	9.184 (0.33)	N	N
Indmixgr of MA 250 k to 500 k	N	18.638** (1.99)	N	15.974 (1.18)	N	-26.325 (-1.42)	N	12.958 (0.96)
Indmixgr of MA 500 k to 1,500 k	N	27.499** (2.38)	N	14.853 (1.11)	N	-36.693 (-1.45)	N	26.755 (1.40)
Indmixgr of MA > 1,500 k	N	61.689** (3.13)	N	4.084 (0.23)	N	-13.440 (-0.61)	N	8.441 (0.41)
Dist x Indmixgr of metropolitan area	N	-0.083 (-0.85)	N	0.420 (0.27)	N	N	N	N
Dist x Indmixgr of MA < 250 k	N	-0.217** (-2.92)	N	0.152 (1.50)	N	-0.055 (-0.28)	N	N
Dist x Indmixgr of MA 250 k to 500 k	N	-0.080 (-1.04)	N	-0.150* (-1.71)	N	0.157 (1.11)	N	-0.011 (-0.11)
Dist x Indmixgr of MA 500 k to 1,500 k	N	-0.198** (-2.46)	N	-0.150* (-1.70)	N	0.246* (1.68)	N	-0.109 (-0.87)
Dist x Indmixgr of MA > 1,500 k	N	-0.376** (-2.64)	N	-0.067 (-0.53)	N	-0.021 (-0.13)	N	-0.030 (-0.25)
R ²	0.60	0.61	0.59	0.63	0.62	0.62	0.62	0.62
No. of counties	1,300	1,300	672	672	416	416	641	641
F-statistic								
Inc distance to MA = 0	12.72**	5.45**	6.21**	3.20**	8.06**	9.71**	0.13	0.28
Indmixgr of MA = 0	N	5.37**	N	11.42**	N	1.27	N	0.97
Dist x Indmixgr of MA = 0	N	4.56**	N	1.86	N	1.04	N	0.27

Notes: t-statistics are in parentheses. They are derived from the Conley (1999) estimator which allows spatial correlation in errors and uses a quadratically declining weighting scheme that becomes zero beyond 200 km. ** and * indicate significant at ≤5% and ≤10% level respectively. N = not included. See the notes to Table 1 for more details.

We now assess the direct role of commuting by going back to the models shown in Table 1 and adding the initial 1990 percent of the county's workforce that is employed in the county of residence – namely, an inverse measure of commuting. As expected, some of the respective distance coefficients are slightly smaller in magnitude (not shown), continuing to illustrate that commuting patterns are not the only reason for the reported distance effects. These results suggest that every one-percentage point increase in the 1990 share of workers employed in the county of residence is associated with between 0.1 to 0.2 percentage point less population growth in the subsequent 1990–2000 period (the variable is significant at the 5% level).

5 Conclusion

Despite the development of many variants of New Economic Geography (NEG), few studies have empirically examined its ability to explain population dynamics in a fairly developed mature urban system. In particular, there has been little investigation of the spatial interactions among urban areas and between the urban core and peripheral regions. Therefore, this study examined US population dynamics across the urban hierarchy, specifically the link between county population growth and geographic proximity to successively higher-tiered urban areas as well as between areas within the same tier of the hierarchy.

We find that rural counties and smaller urban centres have significant positive interactions with their nearest higher-tiered urban areas. We found successive *penalties* in terms of lower growth the farther a rural or smaller urban county was from each higher tier of urban centre. Further analysis suggested that urban job growth stimulated rural population growth in part through commuting opportunities, illustrating the strong forces supporting urban sprawl. We found little evidence consistent with NEG growth shadows, the exception being spatial competition among small MAs.

For counties located in larger MAs, spatial interactions with higher-tiered urban areas were much less evident. We found evidence of urban growth shadows only around the highest-tier MAs, which were being cast on proximate MAs with population between 250,000 and 1.5 million. The general lack of growth shadows suggest that some predictions of NEG and CPT are not particularly germane for describing the continued evolution of the American urban system. Likewise, commuting linkages likely play only a small role in describing interactions for the smallest MAs and no role for describing the interactions between larger MAs. Instead, the evidence is most consistent with lower-ordered places benefiting from closer accessibility in their backward-forward linkages and access to urban amenities. Finally, we found that deconcentration and sprawl remain key features of *intra* urban area settlement patterns for large MAs, *ceteris paribus*.

In terms of policy, these findings have implications for understanding land-use and settlement patterns. Proximity and access to larger urban areas is critical in shaping the size and growth of rural counties and urban areas with less than 1.5 million people. Indeed, these proximity effects are large, appearing to trump other factors such as own size and even amenities. Further analysis for the most recent years revealed that these trends show no signs of abating. Clearly, planners should consider these much broader regional interactions in their development strategies. In addition, to facilitate this process, further research should assess how the distance effects are changing over time and more closely appraise the household and firm-side paths through which these effects are transmitted.

Appendix Table 1. Variable definitions and descriptive statistics (full sample)

Variable	Description	Source	Mean	St. dev.
Population change	Percentage change in population over 1990–2000	1990 2000 Census	11.22	16.00
Dist to nearest/actual urban centre (micropolitan or metropolitan area)	Distance (in km) between centroid of a county and population weighted centroid of the nearest urban centre, if the county is not in an urban centre. It is the distance to the centroid of its own urban centre if the county is a member of an urban centre (in kms).	1990 Census, C–RERL	34.61	32.44
Inc dist to metro	Incremental dist. to the nearest/actual MA in kms	Authors' est.	36.68	49.06
Inc dist to metro >250 k	Incremental distance to the nearest/actual MA with >250,000 population, in 1990 in kms	Authors' est.	56.29	97.27
Inc dist to metro >500 k	Incremental distance to the nearest/actual MA with >500,000 population in 1990 in kms	Authors' est.	40.67	66.83
Inc dist to metro >1,500 k	Incremental distance to the nearest/actual MA with >1,500,000 population in 1990 in kms	Authors' est.	89.77	111.47
Dist to nearest own tier	Distance in kms to the nearest own tier county/urban area. For an urban area, this is the distance from the centre of the urban county to the population–weighted centre of the nearest own–tier urban area.	Authors' est.	42.85	32.18
Population density	1990 county population per square mile	1990 Census	207.83	1,593.40
Nearest/actual urban centre pop	1990 population of the nearest/actual urban centre measured as a MICRO or MA.	Authors' est.	374,271.3	1,377,909.3
Inc pop of nearest metro	Incremental pop. of the nearest/actual MA, 1990	Authors' est.	186,155.0	457,600.8
Inc pop of metro >250 k Inc pop of metro >500 k Inc pop of metro >1,500 k	Incremental population of the nearest/actual MA with >250,000 population; with >500,000 population; or >1.5 million population in 1990.	Authors' est.	n.a.	n.a.
Weather/Amenity	Vector includes: mean January sun hours; mean January temperature (degree F); mean July relative humidity (%); typography score 1 to 24, in which 24 represents the most mountainous; natural amenity rank 1 to 7, with 7 being the highest; % of county area covered by water	ERS, USDA	n.a.	n.a.
Economic/Demographic				
Median HH inc	Median household income 1989	1990 Census	23,842.7	6,388.8
Industry mix growth	Industry mix employment growth, calculated by multiplying each industry's national employment growth (between 1990 and 2000) by the initial period (1990) industry employ. shares in each sector	1990, 2000 BEA, Authors' est.	0.16	0.04
Unemployment rate	1990 Civilian unemployment rate (%)	1990 Census	6.67	3.02
Agriculture share	1990 Percent employed in agriculture sector	1990 Census	8.45	8.20
Goods share	1990 Percent empl. in (nonfarm) goods sector	1990 Census	27.28	10.19

Appendix Table 1. Continued

Variable	Description	Source	Mean	St. dev.
Age Shares	Percent of 1990 population <6 years; 7–17 years; 18–24 years; 55–59 years; 60–64 years; and >65 years.	1990 Census	n.a.	n.a.
Educational Attainment	% of 1990 population 25 years and over that are high school graduates; have some college; have an associate degree; and are 4 year college graduates.	1990 Census	n.a.	n.a.
Race/Ethnic	% of 1990 population Hispanic; Black; Asian and Pacific Islands; Native American; other race.	1990 Census	n.a.	n.a.
Percent immig 1985–90	Percent of 1990 pop. immigrated over 1985–90	1990 Census	0.48	0.96
Surrounding Variables				
Population density_surr	Weighted average population density in surrounding counties within a BEA region ^a	1990 Census, Authors' est.	663.44	1,553.27
Median HH inc_surr	Weighted average median household income in surrounding counties within a BEA region ^a	1990 Census, Authors' est.	26,753.7	4795.7
Industry mix growth_surr	Weighted average industry mix employment growth in surrounding counties within a BEA region ^a	1990 BEA, Authors' est.	0.19	0.02
Unemployment rate_surr	Weighted average total civilian unemployment rate in surrounding counties within a BEA region ^a	1990 Census, Authors' est.	6.25	1.55
No. of counties			3,029	

Notes: Centroids are population weighted. The metropolitan/micropolitan definitions follow from the 2003 definitions. BEA = Bureau of Economic Analysis, Regional Economic Information Service; ERS, USDA = Economic Research Services, U.S. Department of Agriculture; C–RERL = Canada Rural Economy Research Lab, University of Saskatchewan. See Partridge and Rickman (2006) for more details of the variable sources and sample selection.

^a The surrounding BEA region variables are calculated as the average of the region net of the county in question. The BEA economic regions are 177 functional economic areas constructed by the BEA.

Appendix Table 2. Mean (standard deviations) of major variables by population group

Variables	Noncore rural	Micropolitan	MA ≤ 250,000	MA > 250,000
Population change	6.17 (13.44)	10.21 (13.18)	15.69 (15.20)	19.62 (19.39)
Distance to nearest or actual urban center	59.91 (30.56)	4.63 (9.63)	17.76 (18.60)	28.60 (19.52)
Inc dist to MA	43.47 (49.93)	78.46 (46.97)	n.a.	n.a.
Inc dist to MA > 250,000	76.02 (115.19)	48.96 (83.41)	93.23 (93.26)	n.a.
Inc dist to MA > 500,000	45.32 (68.95)	38.17 (59.87)	36.87 (59.07)	36.29 (73.34)
Inc dist to MA > 1,500,000	83.45 (106.24)	99.81 (119.29)	78.54 (115.44)	99.37 (139.88)
Dist to nearest own tier	42.60 (22.43)	50.26 (34.47)	45.99 (44.38)	33.53 (34.69)
Population density	23.01 (20.06)	61.14 (45.95)	119.96 (123.82)	793.46 (3,399.82)
No. of counties	1,300	672	416	641

Notes: The categories are determined using 2003 definitions.

Appendix Table 3. Dependent variable: 1990–2006 %Δ in US county population

	Noncore Rural Areas	Micropolitan Areas	Small MAs	Large MAs
Intercept	8.628 (0.31)	4.609 (0.12)	−12.677 (−0.24)	65.602 (1.45)
Distance to nearest or actual urban center	−0.150** (−7.85)	0.200** (2.13)	0.107 (1.28)	0.303** (3.61)
Inc Dist to MA	−0.053** (−4.82)	−0.039** (−2.05)	n.a.	n.a.
Inc Dist to MA > 250,000	−0.041** (−4.72)	−0.037** (−2.98)	−0.095** (−5.16)	n.a.
Inc Dist to MA > 500,000	−0.027** (−2.28)	−0.021 (−1.26)	−0.050** (−2.46)	0.015 (0.53)
Inc Dist to MA > 1,500,000	−0.015** (−2.45)	0.006 (0.63)	−0.060** (−3.27)	−0.008 (−0.55)
Dist to nearest own tier	−0.014 (−0.53)	0.034 (1.25)	0.065** (2.63)	0.013 (0.36)
Population density	−0.014 (−0.30)	−0.027 (−1.22)	−0.037** (−3.85)	0.001** (2.30)
Pop of nearest or actual urban center	1.2E−05* (1.93)	−3.2E−05 (−0.98)	7.4E−05** (3.13)	−9.3E−07 (−1.37)
Inc pop of nearest MA	7.5E−07 (0.61)	−3.8E−07 (−0.28)	n.a.	n.a.
Inc pop of MA > 250 k	−5.1E−08 (−0.05)	−6.3E−07 (−0.63)	1.4E−06 (1.64)	n.a.
Inc pop of MA > 500 k	−4.4E−07 (−0.67)	−2.1E−07 (−0.35)	−5.0E−07 (−0.66)	−9.1E−07 (−1.44)
Inc pop of MA > 1,500 k	−2.8E−07 (−0.90)	−3.0E−07 (−1.13)	4.9E−07 (0.78)	−8.6E−07** (−2.25)
Pop in surrounding counties	−4.8E−07 (−1.23)	1.9E−07 (0.25)	−8.4E−07 (−1.41)	5.5E−07 (1.20)
Weather/Amenity ^a	Y	Y	Y	Y
Economic/Demographic ^b	Y	Y	Y	Y
Surrounding Econ/Demog ^c	Y	Y	Y	Y
State fixed effects (FE)	Y	Y	Y	Y
R ²	0.61	0.61	0.61	0.60
No. of counties	1,300	672	416	641
F-statistic				
All MA pop = 0	2.43**	0.31	2.26*	0.94
Inc MA pop = 0	0.46	0.17	0.92	1.39
Inc distance to MA = 0	10.85**	3.14**	6.99**	0.43

Notes: Robust t-statistics from Conley (1999) estimator are in the parentheses.

A ** or * indicates significant at ≤5% or ≤10% level respectively.

N = not included, Y = included.

^a sunshine hours, January temp, July humidity, typography, amenity ranking, and percent water area.

^b 1989 median household income, 1990–2000 industry mix emp. growth, 1990 unemp. rate, 1990 share ag. emp., 1990 share goods emp., 6 age-distribution variables for 1990, 4 education categories for 1990, 5 race/ethnicity variables for 1990, and percentage of population immigrated during 1985–1990.

^c weighted average 1989 median household income, 1990–2000 industry mix emp. growth, and 1990 unemp. rate in surrounding counties within a BEA region.

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¿Son las sombras de aglomeración de la Nueva Geografía Económica la raíz de las dinámicas de población actuales de la jerarquía urbana?

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Resumen. Aunque la Nueva Geografía Económica (NEG, por sus siglas en inglés) ha sido utilizada ampliamente para explicar de modo formal el surgimiento del sistema urbano estadounidense, pocos estudios investigan su éxito en explicar las dinámicas de población actuales en un sistema urbano más establecido. Este estudio explora la posibilidad de que centros urbanos del mismo tamaño y de elevada magnitud afectó a los patrones de crecimiento poblacional en condados de los EE.UU. en 1990-2006. En lugar de proyectar sombras de aglomeración de acuerdo a la NEG sobre el crecimiento cercano, los centros urbanos más grandes parecen tener en general efectos de crecimiento positivos en lugares cercanos de menos de 250,000 habitantes. Sin embargo, existen indicios de que las áreas urbanas más grandes proyectan sombras de crecimiento sobre áreas metropolitanas próximas de tamaño mediano y de competencia espacial entre áreas metropolitanas pequeñas.

JEL classification: R11, R12

Palabras clave: Nueva Geografía Económica, sombra de aglomeración, crecimiento de población, jerarquía urbana

要旨： 新経済地理学 (New Economic Geography : NEG)は米国の都市システムの出現を公式に説明するために広範に用いられてきたにもかかわらず、より確立した都市システムにおける現在の人口動態の説明において新経済地理学の成功を吟味した研究はほとんどなされていない。本研究では、同規模でより階層化が進展している都心部への近接性が、1990年から2006年にかけての米国の人口増加パターンに影響を及ぼしたか否かを検証する。新経済地理学における集積の影は近隣地域に影響を及ぼすものではなく、より規模の大きい都心部は一般に人口25万人未満の隣接地域にプラスの影響を及ぼすことが明らかになった。しかし、最大規模の都市部は、近接する中規模都市部及び小規模都市部間の空間的競争に成長の影を落とすという証拠がいくつかある。